

Class

- A template for creating objects that bundles data (attributes) and behavior (methods) together.
- define as `class ClassName:`
- the `__init__` method sets up the initial attributes when an object is created
- Each object made from the class is called an instance
- `self` refers to that specific instance inside the class's methods, letting each object keep its own data while sharing the same behavior.

```
In [... s = 'string'
s.capitalize()
a = [1,3,4]
print(type(a))
```

```
Out[... 'String'
<class 'list'>
```

```
In [... class Person:
    def __init__(self, a):
        self.name = a

    def find_name_length(self):
        print(f'The length of {self.name} is {len(self.name)}')

p1 = Person('debjit')
print(p1.name)
p1.find_name_length()
```

```
debjit
The length of debjit is 6
```

```
In [... class Customer:
    def __init__(self, name, age, current_balance=0):
        self.name = name
        self.age = age
        self.current_balance = current_balance
```

```
def change_balance(self, x):  
    self.current_balance = self.current_balance +
```

```
In [... person1 = Customer("Alice", 25)  
  
# person1.__dir__()  
person1.name  
person1.age  
person1.current_balance  
deposit = -10 # real banks will do something more dec  
person1.change_balance(deposit)  
person1.current_balance
```

```
Out[... 'Alice'
```

```
Out[... 25
```

```
Out[... 0
```

```
Out[... -10
```

```
In [... person2 = Customer("Bob", 20, 56)  
# type(person2)  
person2.name  
person2.age  
person2.current_balance  
deposit = 10 # real banks will do something more dece  
person2.change_balance(deposit)  
person2.current_balance
```

```
Out[... 'Bob'
```

```
Out[... 20
```

```
Out[... 56
```

```
Out[... 66
```

```
In [... class Student:  
    def __init__(self, name, grade):  
        self.name = name  
        self.grade = grade  
  
    def is_passing(self):  
        return self.grade >= 50
```

```
In [... student = Student("John", 5)
```

```
print(student.name)  
print(student.grade)  
print(student.is_passing())
```

John

5

False